

A machine learning based 4-prong tagger for hadronic decays of highly boosted WW in ATLAS

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On behalf of the ATLAS Collaboration



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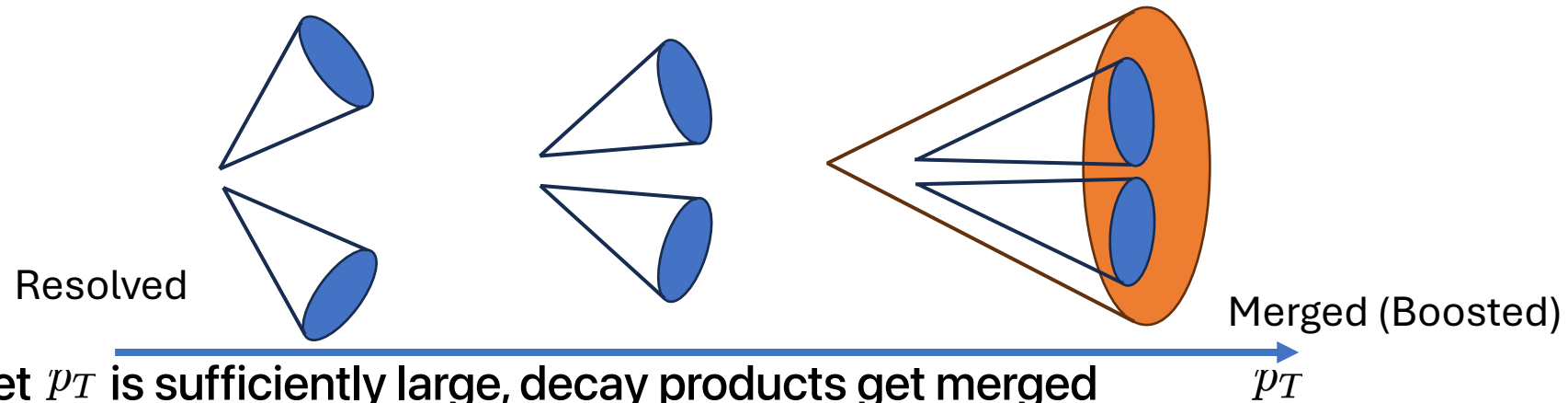
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Jets in ATLAS

(See talk by Matthew Green)

- Hadronic signatures reconstructed as small-radius ($R = 0.4$) [anti-kt](#) jets

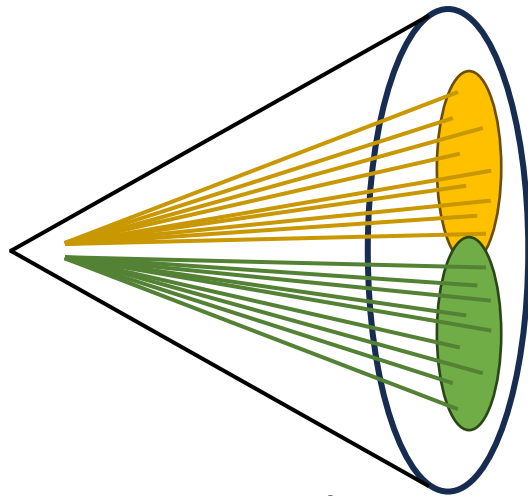


- When the jet p_T is sufficiently large, decay products get merged
 - Capture the decay products as a fat jet
 - Large-radius ($R=1.0$) anti-kt jet
 - [Unified-Flow Object](#) constituents: Calorimeter + Tracker combined object
 - Use its substructure for discrimination - "Boosted objects tagging"
 - Machine learning to further improve the performance

4-prong jet

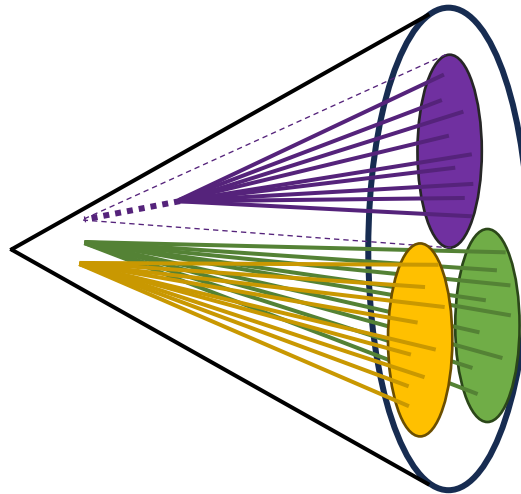
([See talk by Robert Les](#))

- Various boosted object identification in ATLAS
 - Prong: Signature initiated by a single quark
- Constituent based jet tagger approach



2-prong Jet

[ATL-PHYS-PUB-2023-020](#)



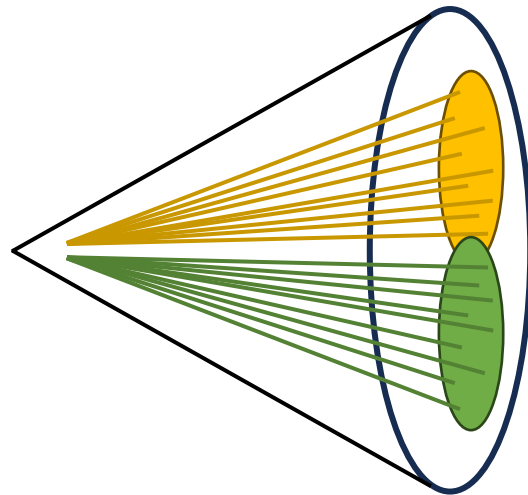
3-prong Jet

[JETM-2026-03](#)

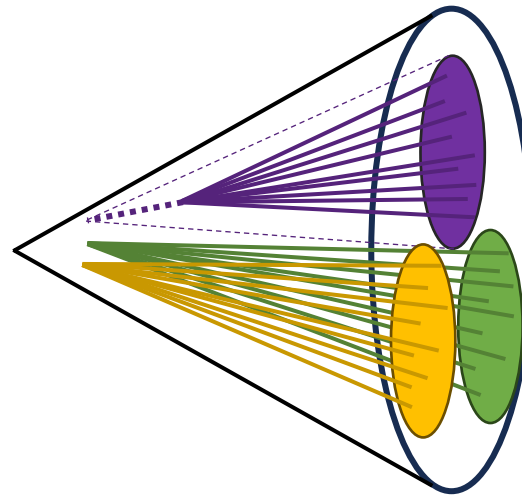
4-prong jet

([See talk by Robert Les](#))

- Various boosted object identification in ATLAS
 - Prong: Signature initiated by a single quark
- Constituent based jet tagger approach
- 4-prong jet tagger: First time in ATLAS!

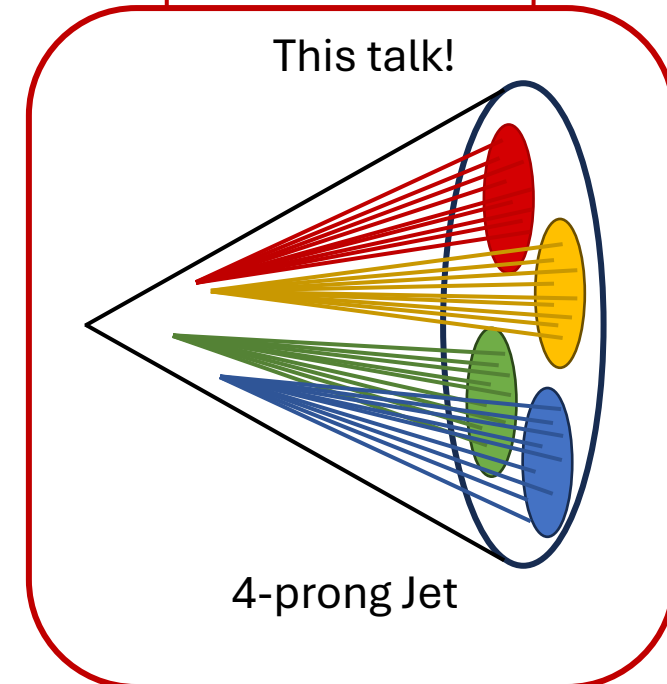
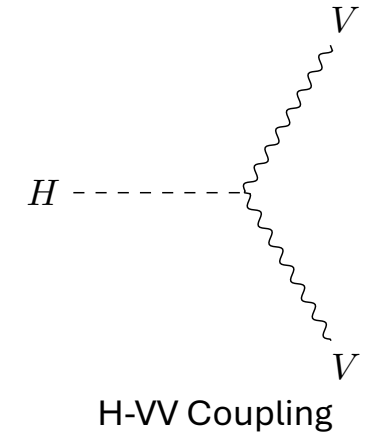


2-prong Jet
[ATL-PHYS-PUB-2023-020](#)



3-prong Jet
[JETM-2026-03](#)

Search for new physics
in 4-prong topology

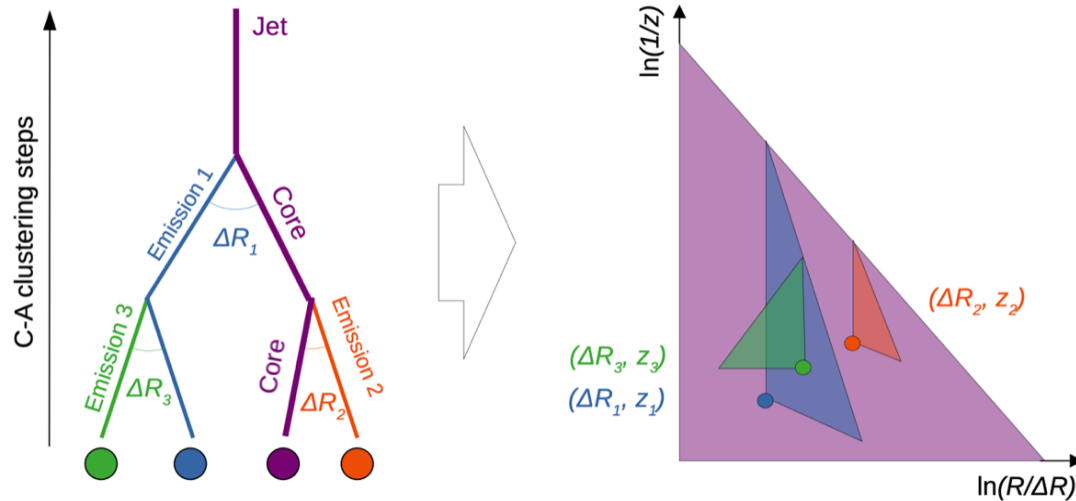


This talk!

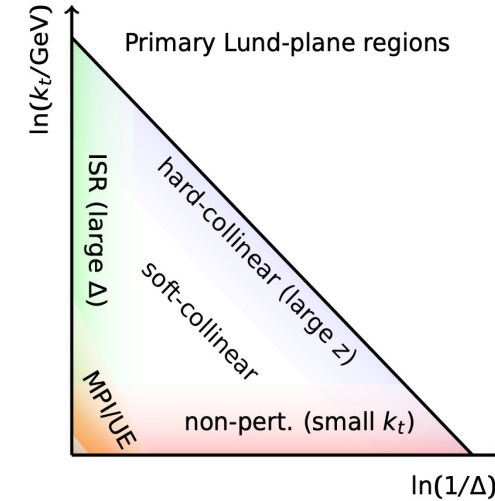
4-prong Jet

Lund jet plane

- One of the substructure representation
- Lund jet plane constructed by the Cambridge-Aachen (C/A) algorithm
 - Follows the clustering history
- Different regions represents distinct physical regimes inside jet



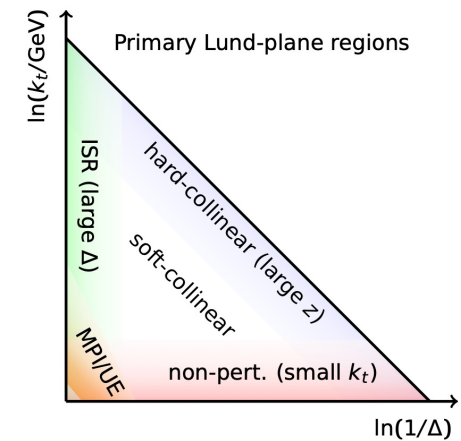
[[ATL-PHYS-PUB-2023-017](#)]



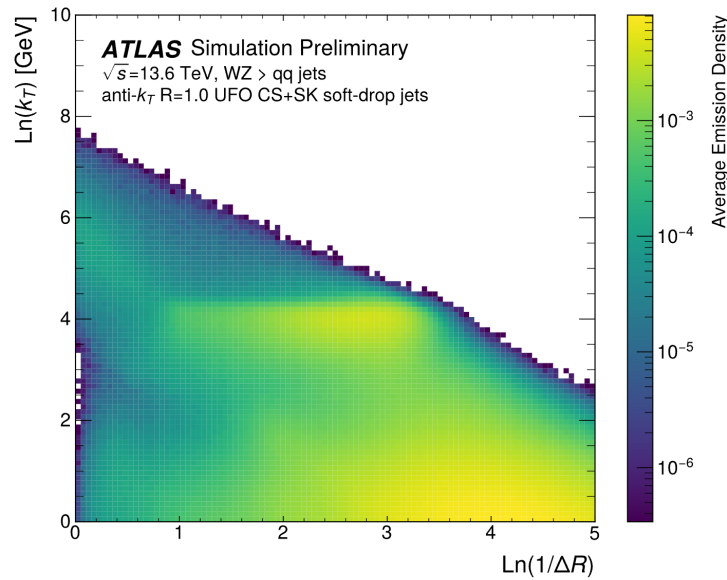
The Lund jet plane [[10.1007/JHEP12\(2018\)064](#)]

Lund jet plane

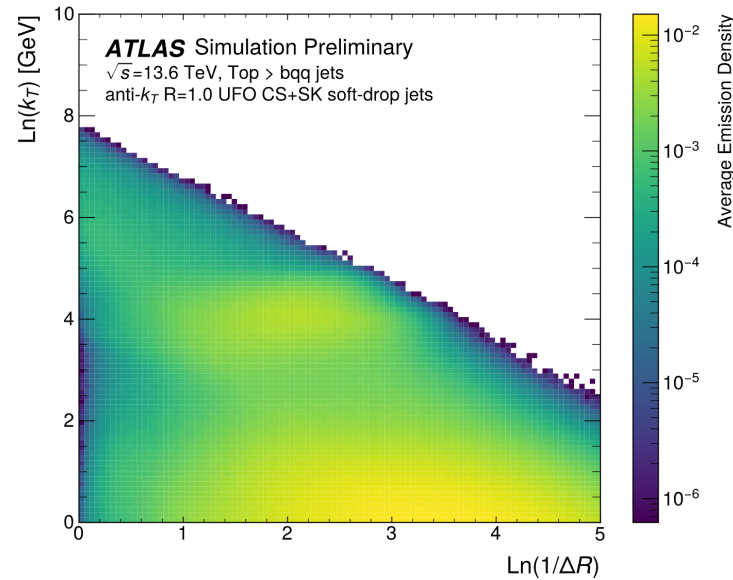
- Lund jet planes of various large-R jet topologies



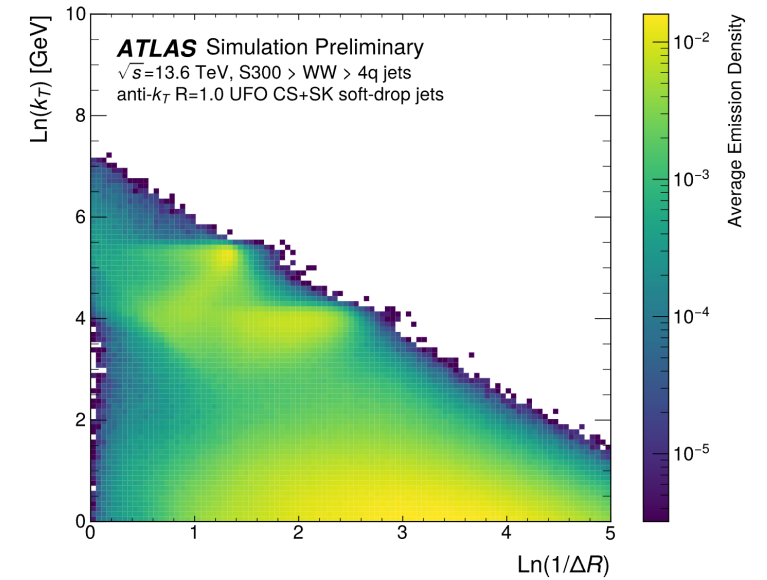
The Lund jet plane [[10.1007/JHEP12\(2018\)064](https://arxiv.org/abs/10.1007/JHEP12(2018)064)]



2-prong Jet ($W/Z \rightarrow qq$)



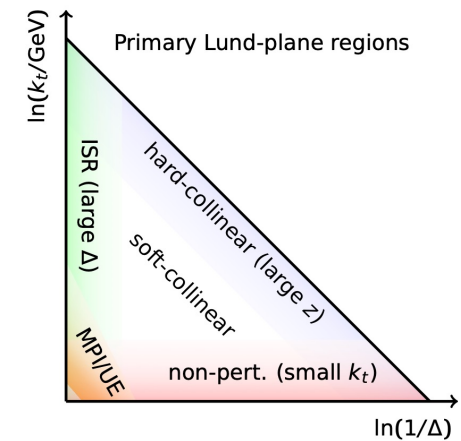
3-prong Jet (Top $\rightarrow bqq$)



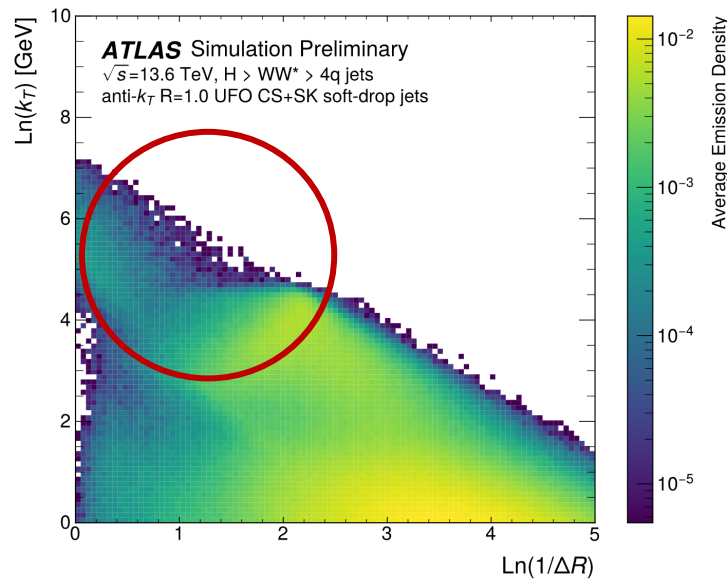
4-prong Jet ($S \rightarrow WW \rightarrow 4q$)

Lund jet plane

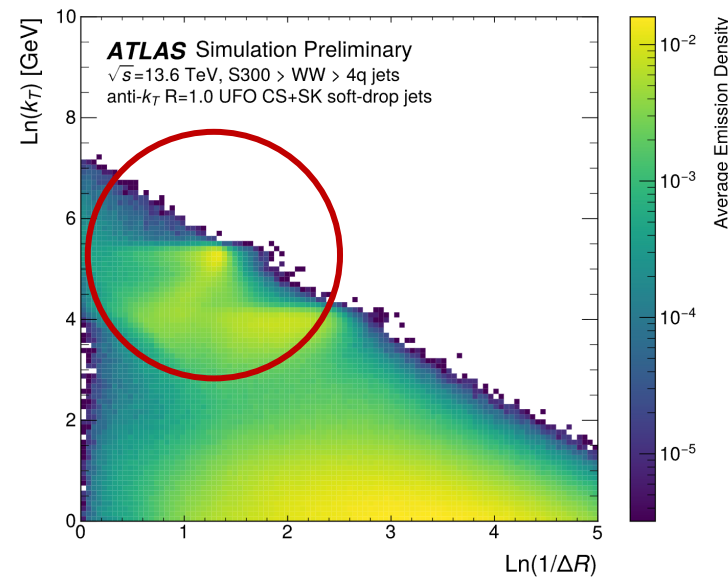
- Lund jet planes of various large-R jet topologies
- 4-prong jet: Upper left distribution depends on the scalar particle's mass



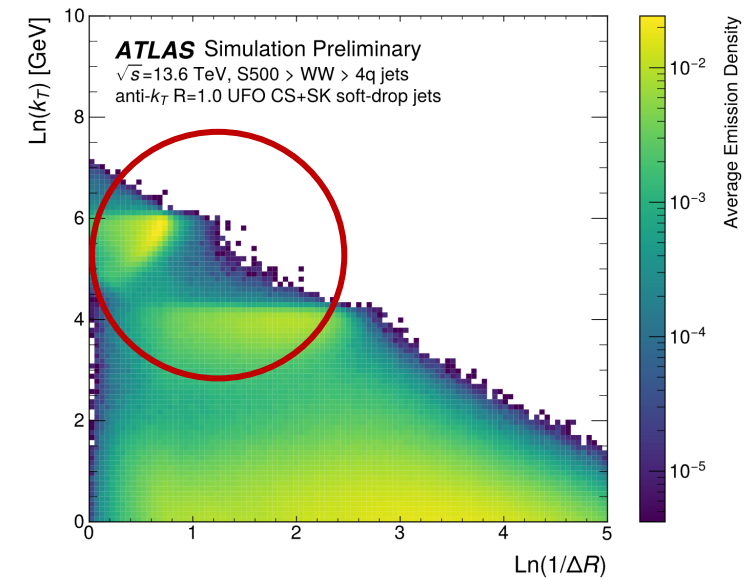
The Lund jet plane [[10.1007/JHEP12\(2018\)064](https://arxiv.org/abs/10.1007/JHEP12(2018)064)]



125 GeV ($H \rightarrow WW^* \rightarrow 4q$)



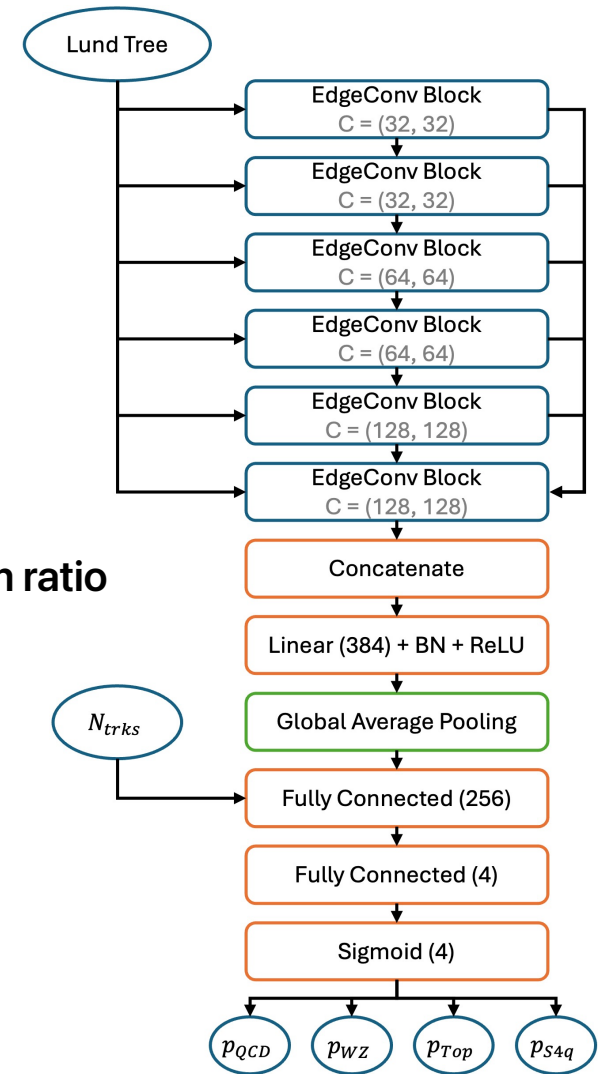
300 GeV ($S \rightarrow WW \rightarrow 4q$)



500 GeV ($S \rightarrow WW \rightarrow 4q$)

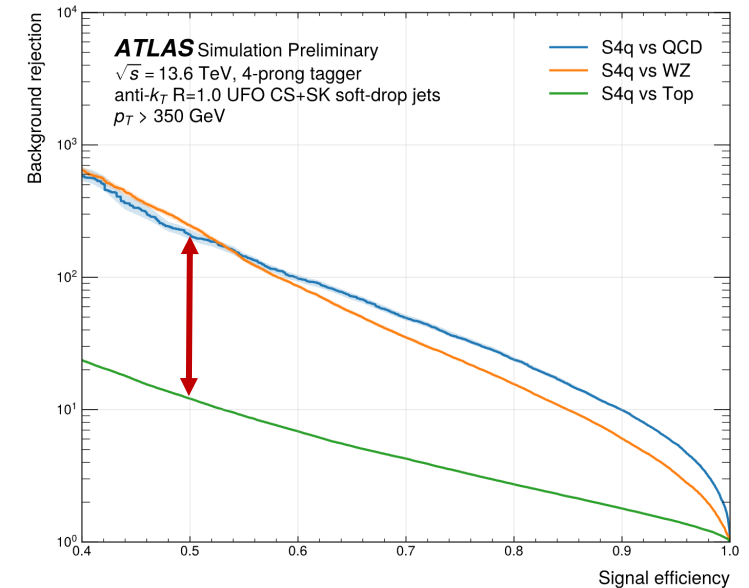
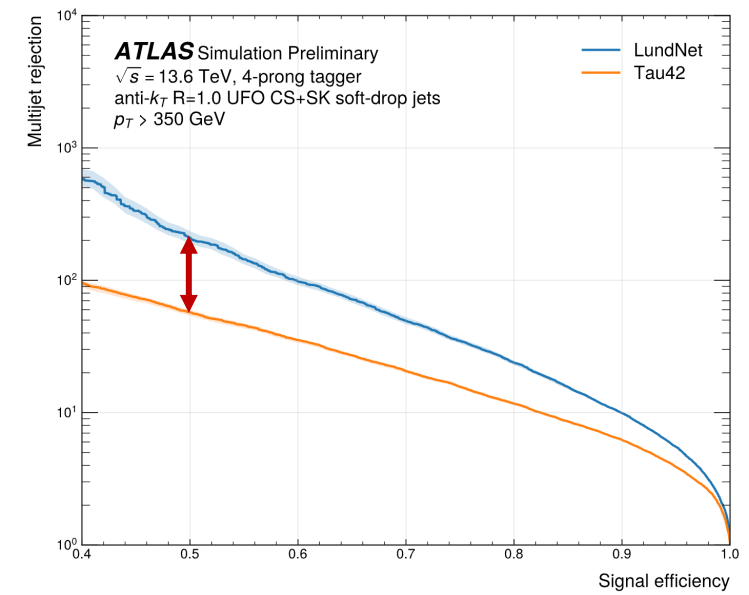
LundNet

- LundNet is a Graph-Neural-Network used for jet tagger
- ATLAS presented W-tagger previously [[ATL-PHYS-PUB-2023-017](#)]
- Binary tree input from the C/A clustering history
- LundNet takes three node features
 - $\ln(1/\Delta R)$, $\ln(k_T)$, $\ln(z)$ - Split angle, transverse momentum, momentum ratio
- Number of tracks as global feature
- Four output classes – 1M Jets per class
 - 1-prong: QCD multijet process (q/g)
 - 2-prong: Vector-boson ($W/Z \rightarrow qq$)
 - 3-prong: Top-quark ($\text{Top} \rightarrow bqq$)
 - 4-prong: Scalar $\rightarrow WW \rightarrow 4q$
 - On-shell W-boson only



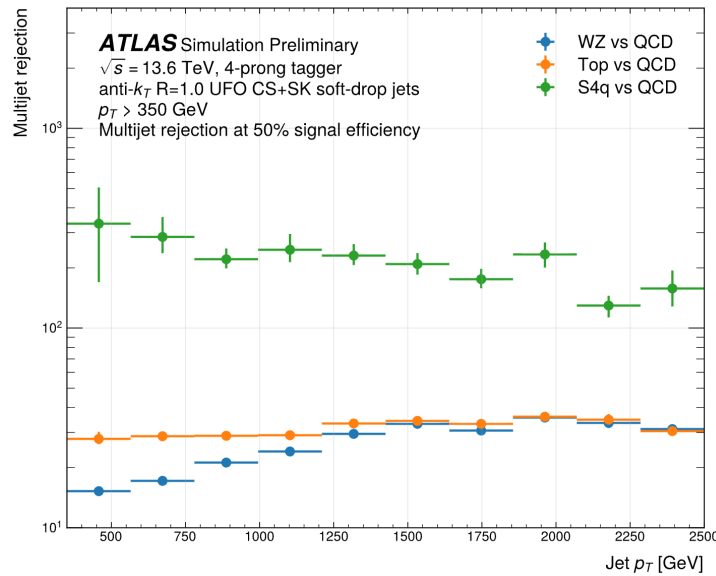
Performance

- 4-prong jet signal identification performance
 - Signal: $m_S = 300$ GeV initiated 4-prong jet
- LundNet vs τ_{42} ratio
 - τ_{42} : four subjettness ratio
 - Background: QCD Multijet process
 - LundNet outperforms τ_{42} by 4x
 - At 50% signal efficiency point
- 4-prong signal vs various backgrounds
 - Effectively reject QCD and WZ classes
 - Not as performant as for Top background
 - Similar 3-prong vs 4-prong topologies
 - No flavour information

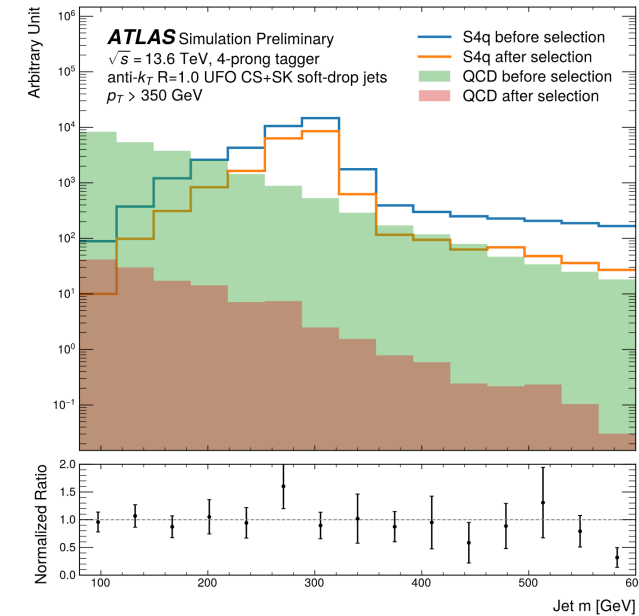


Performance

- No mass constraint on the BSM scalar S
 - Mass agnostic tagger is favored
- pT and mass decorrelation by reweighting the training dataset



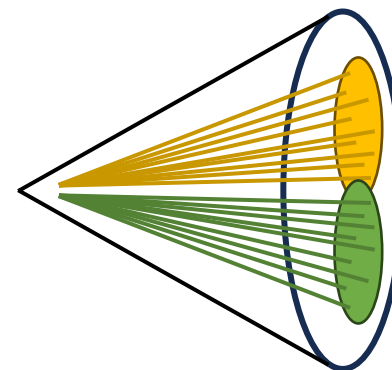
QCD rejection vs Large-R jet pT



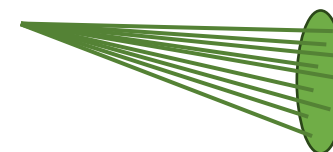
QCD mass before and after selection

Per-prong correction method

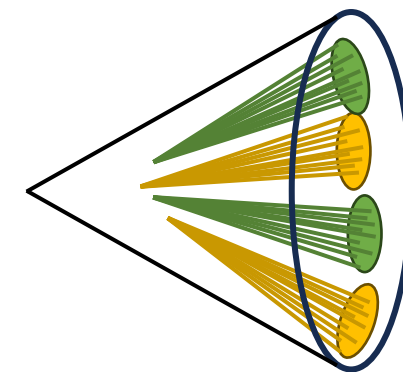
- Main challenge: 4-prong tagger calibration
 - There are no SM proxy 4-prong jets in data
 - How do we estimate the 4-prong ε_{Data} ?
- Recently, the CMS Collaboration presented a novel 4-prong calibration method
 - [CMS PAS JME-23-001](#)
- ATLAS started exploring this new technique as well
 - Lund jet planes of subjets are identical for all n-prong topologies
 - Applicable to any number of prongs
 - Use subjets of the well-known 2-prong and 3-prong jets
 - Lund jet plane $\rightarrow$ subjet $\rightarrow$ jet correction (Bottom-up approach)



Identify known topology



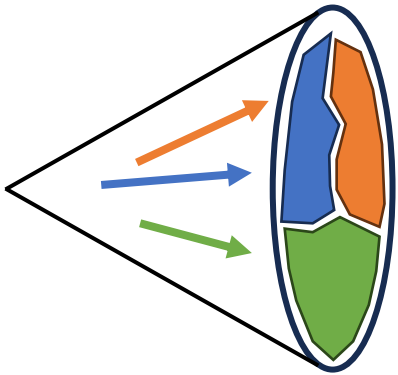
Subjet substructure (Lund jet plane)



Apply to n-prong topology

Per-prong correction method

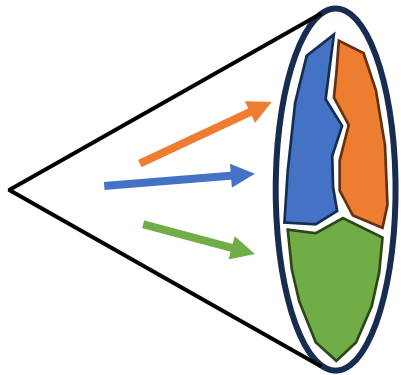
- Methodology
 - Identify subjects inside large-radius jet using the C/A algorithm



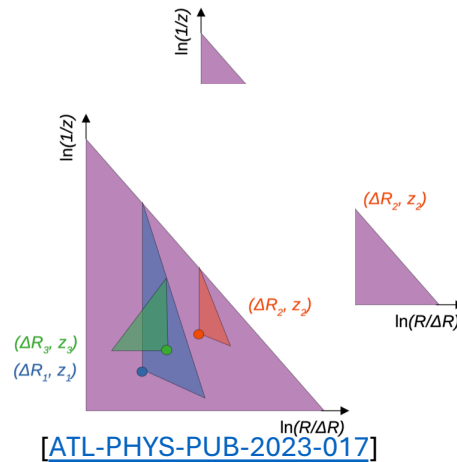
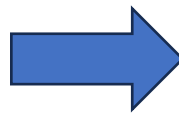
Exclusive subject reclustering

Per-prong correction method

- Methodology
 - Identify subjects inside large-radius jet using the C/A algorithm
 - For each subject create Lund jet plane using the C/A algorithm



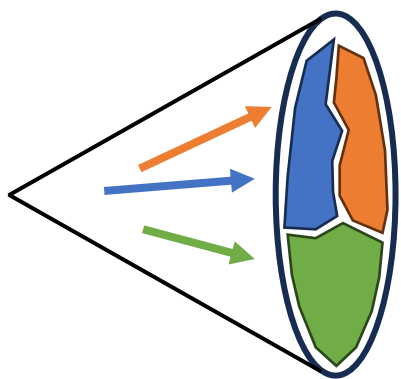
Exclusive subject reclustering



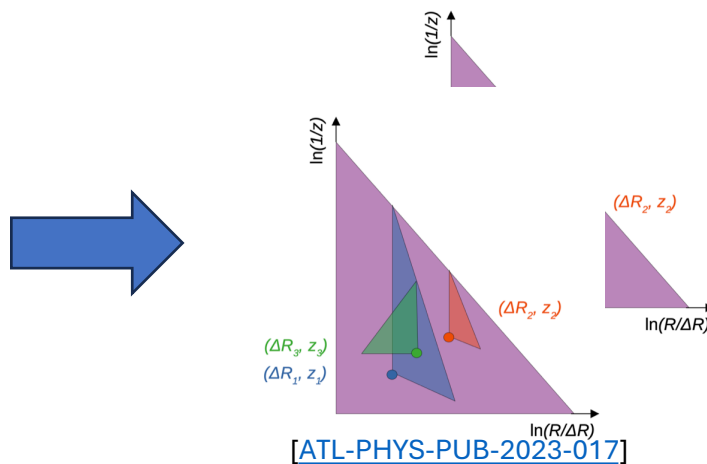
Subject Lund jet plane per dataset

Per-prong correction method

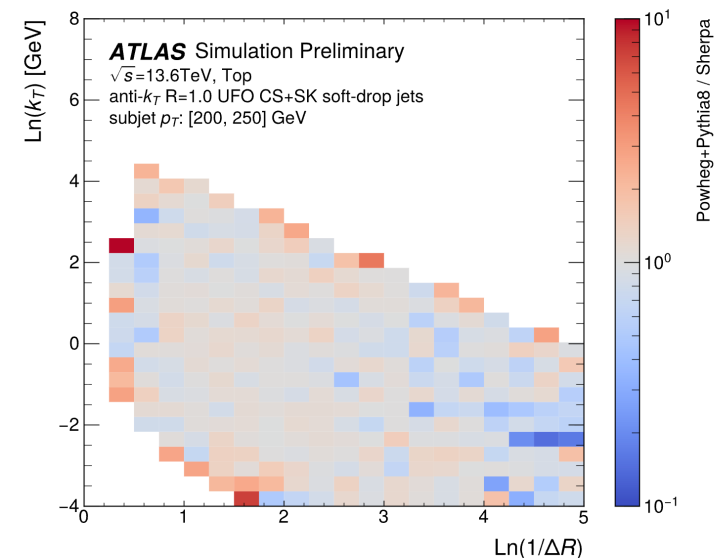
- Methodology
 - Identify subjects inside large-radius jet using the C/A algorithm
 - For each subject create Lund jet plane using the C/A algorithm
 - Derive the subject Lund jet plane density ratio between two datasets



Exclusive subject reclustering



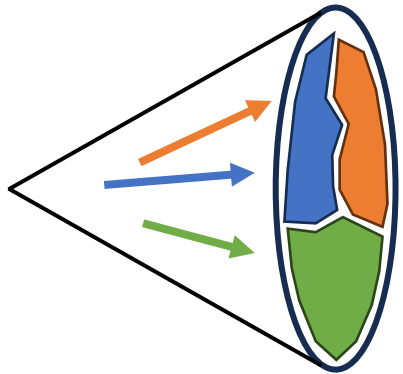
Subject Lund jet plane per dataset



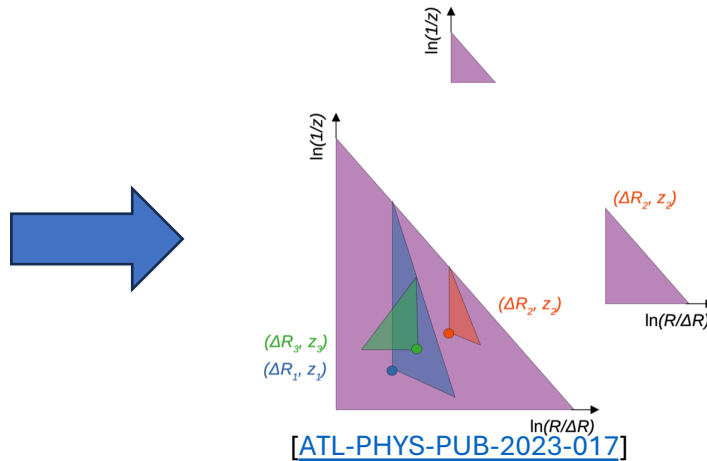
Lund jet plane density ratio (Correction factor)

Per-prong correction method

- Methodology
 - Identify subjects inside large-radius jet using the C/A algorithm
 - For each subject create Lund jet plane using the C/A algorithm
 - Derive the subjet Lund jet plane density ratio between two datasets
 - Multiply the correction factor (CF) for each C/A split



Exclusive subjet reclustering



Subject Lund jet plane per dataset

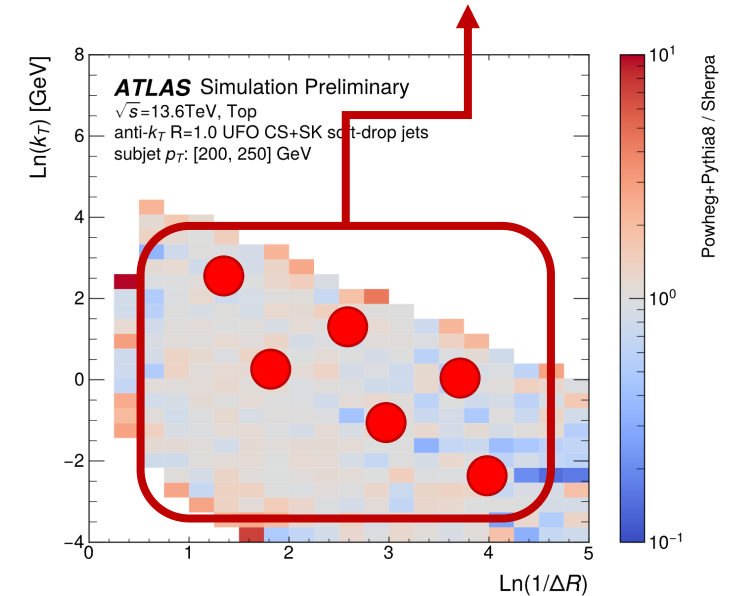
[ATL-PHYS-PUB-2023-017]

Multiply the correction factor as weight

$$CF_{\text{Subjet } j} = \prod_{i \in \text{C/A splits}} CF_i$$

$$CF_{\text{Jet}} = \prod_j CF_{\text{Subjet } j}$$

$$= \prod_j \left[\prod_{i \in \text{C/A splits}} CF_i \right]_j$$

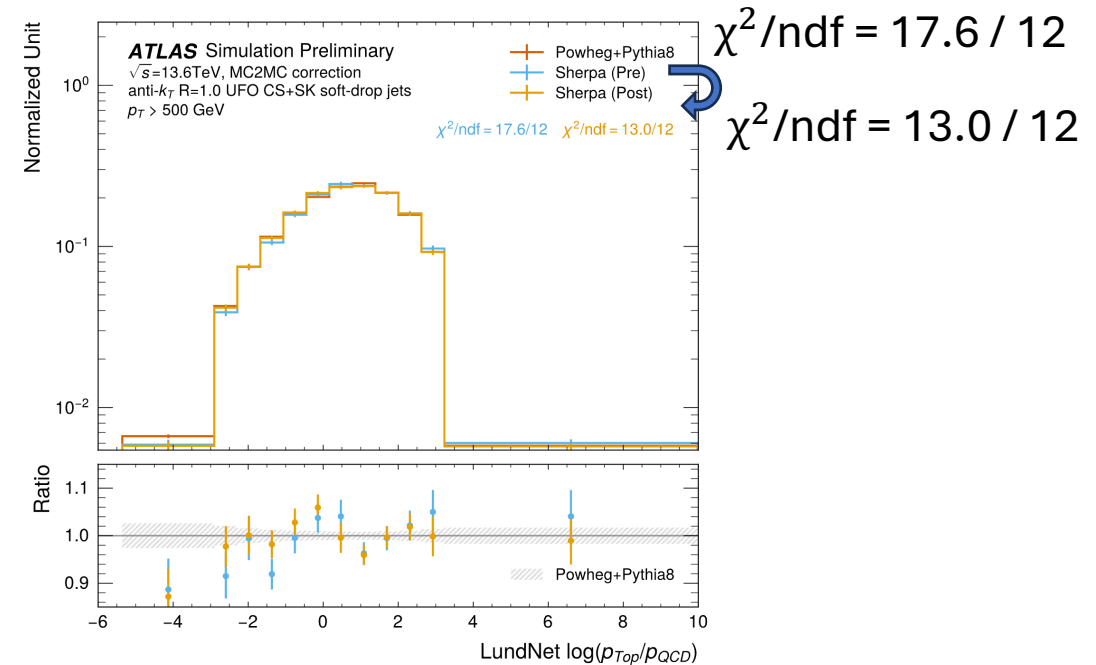
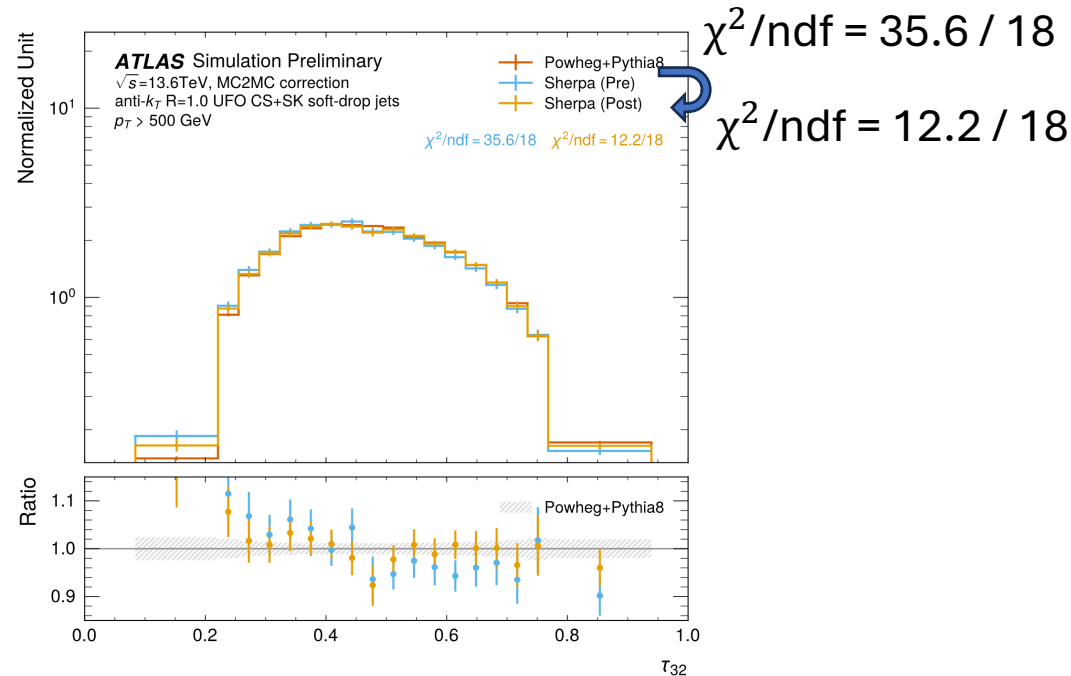


Lund jet plane density ratio (Correction factor)

Powheg+Pythia8 / Sherpa

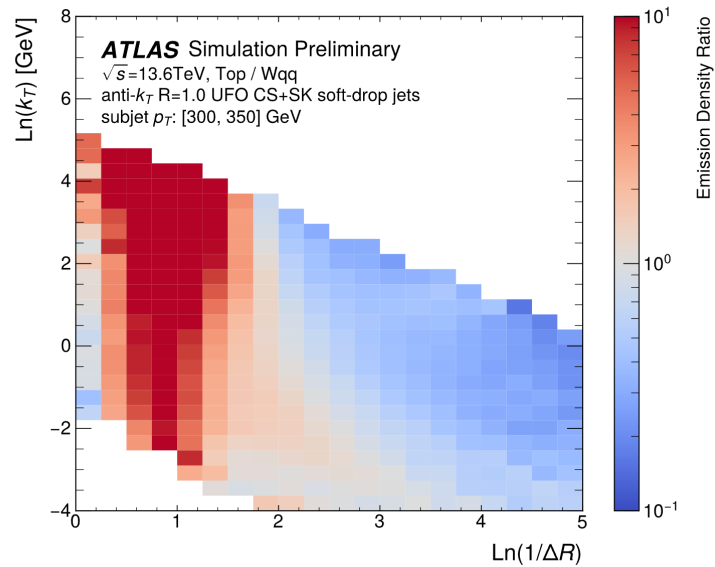
Per-prong correction method

- Proof-of-concept: MC2MC correction using ttbar samples
 - Apply correction from systematic (Sherpa) to the nominal (PowhegPY8)
 - Compare normalized distribution of τ_{32} and LundNet 3-prong discriminator

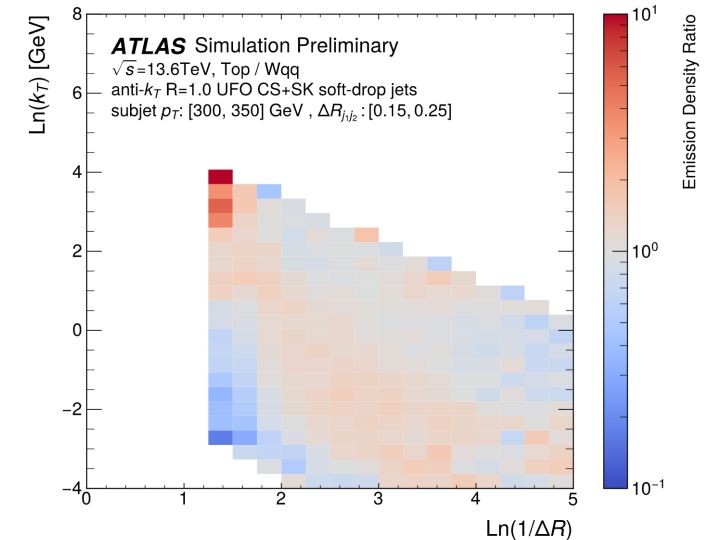
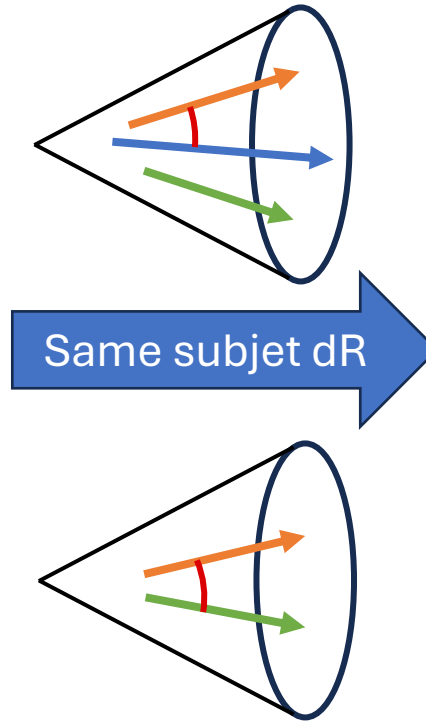


Per-prong correction method

- Possible sources of systematics – Are the Lund jet plane identical for different n-prong jets?
- Subjet Lund jet plane between 2-prong vs 3-prong jets
- Top / W subjet Lund jet plane ratio



Inclusive Lund jet plane density ratio



Lund jet plane density ratio
with subjet dR selection

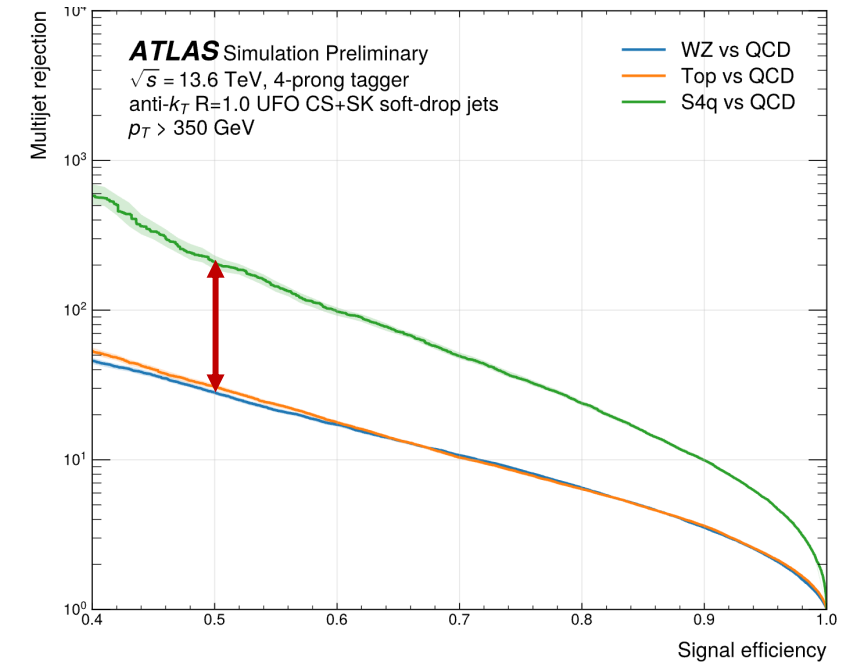
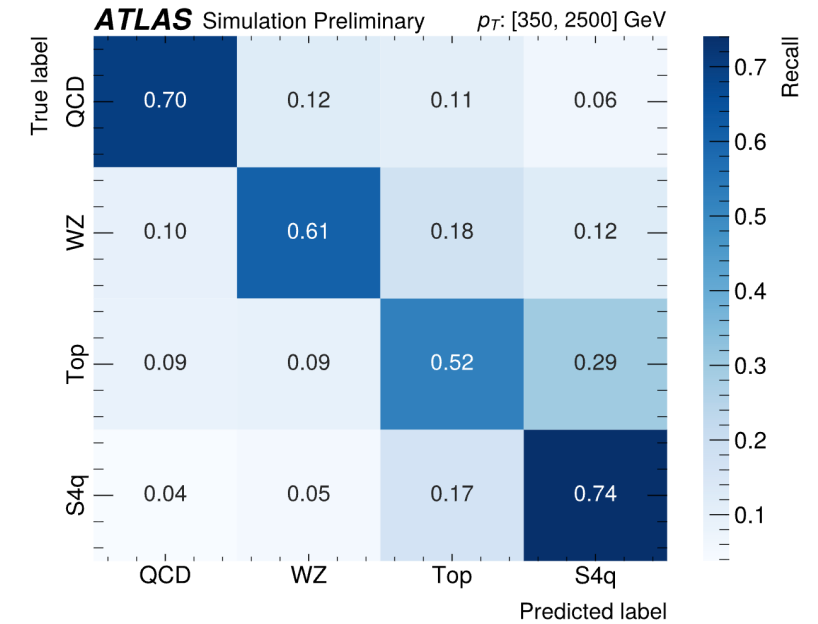
Summary

- Boosted 4-prong jet tagger developed by ATLAS for the first time
- Enhanced QCD multijet rejection with respect to τ_{42} ratio
- Per-prong correction method first explored by ATLAS
- Stay tuned for more on the 4-prong tagging and 4-prong calibration from ATLAS
- Thank you for your attention!

Backup

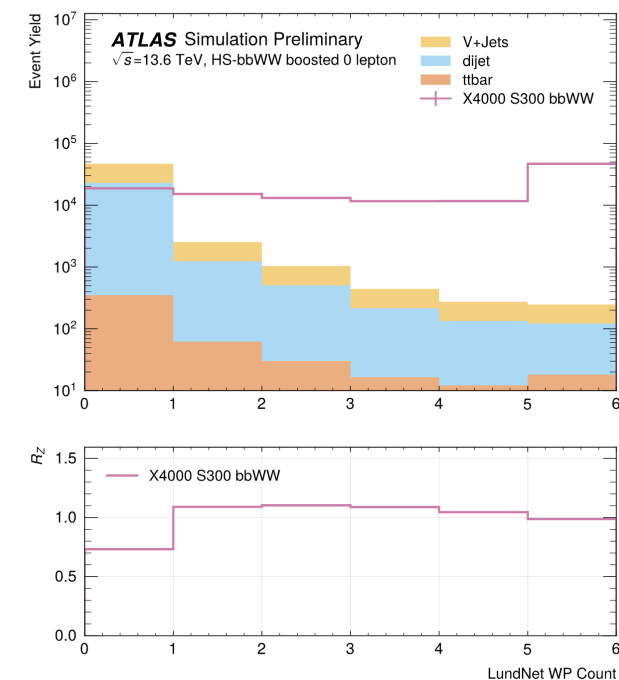
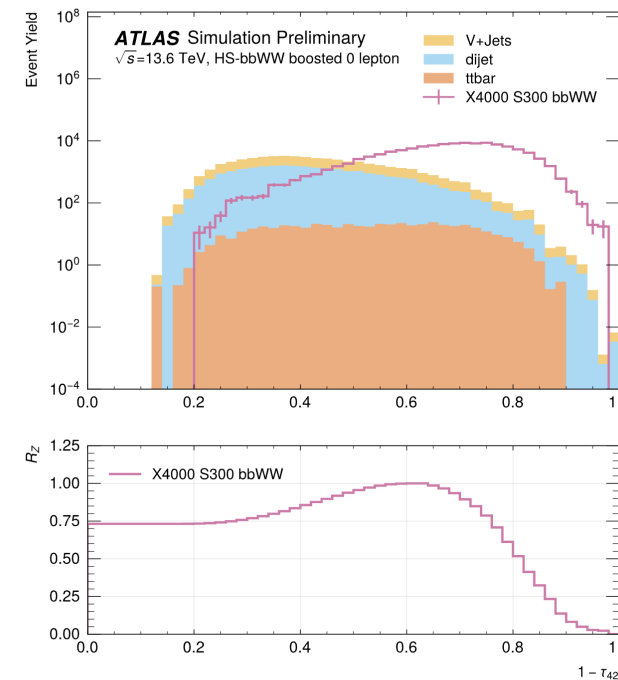
The 4-prong LundNet

- Confusion matrix
 - High top vs 4-prong mis-tag rate
 - Difficult to discriminate 3-prong vs 4-prong
 - Without any track information
 - Room for improvement for top class
- Various signal vs QCD backgrounds
 - Generic n-prong tagger
 - Identifying 4-prong is easier than 2-prong, 3-prong jets



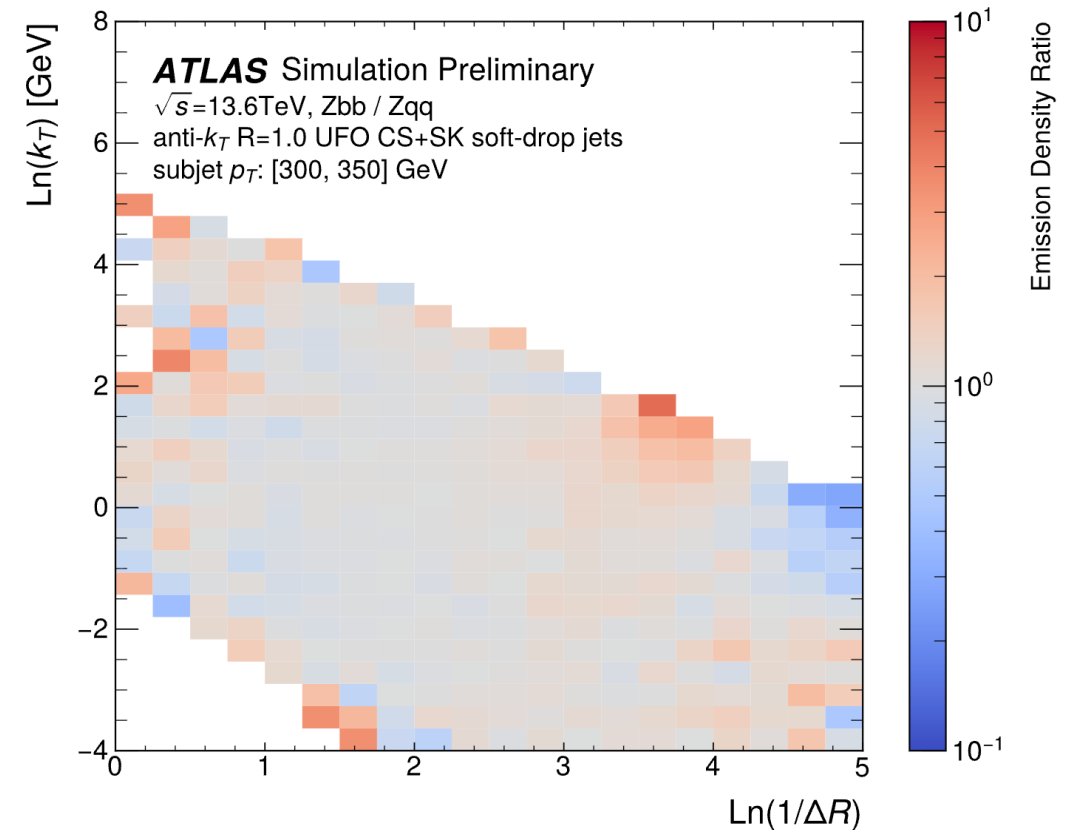
The 4-prong LundNet

- Toy analysis example – $X \rightarrow HS \rightarrow bbWW$
 - 2 large-R jets
 - $H \rightarrow bb$ (Hbb)
 - $S \rightarrow WW \rightarrow 4q$ (S4q) : The 4-prong jet
- Compare two 4-prong selection strategies
 - τ_{42} cut based selection
 - LundNet WP Count
 - Number of WP a jet has passed
 - Pseudo-continuous [50, 60, 70, 80, 90]
- Compare relative significance
 - Baseline: τ_{42} threshold @ 0.4
 - LundNet: LundNet WP @ 2
 - 10% improved significance at WP 80



Per-prong correction method

- Possible sources of systematics
- Subjet Lund jet plane flavour dependence
 - b-subjet vs q-subjet
- Zbb / Zqq subjet Lund jet plane ratio
- Smaller difference compared to Slide 15
- Difference observed at small dR region
- Further study is needed



Lund jet plane density ratio: b-subjet / l-subjet